Unique paper code: 6332493501
Name of Paper: Full Stack Web Development-2
Name of the Course: B.Voc. (Software Development)
Semester: 5
Duration: 3 Hours
Maximum Marks: 90

Instructions for Candidates:

- **Write your Roll No. on the top immediately on receipt of the question paper.**
- **The paper has two sections. Section A is compulsory. Each question is of 5 marks.**
- **Attempt any four questions from Section B. Each question is of 15 marks.**

Section A

1 a) What is `Node.js`? Explain its main features and advantages. 5

b) Differentiate between `fs.readFile()` and `fs.readFileSync()` in `Node.js`, and provide an example demonstrating their usage. 5

c) Explain how request handling works in a `Node.js` HTTP server. Illustrate the process with a diagram showing the request flow. 5

d) What is the difference between `local` and `global` packages in `Node.js?` 5

e) List and explain any three key elements of a `pom.xml` file. 5

f) What is `Express.js`? Describe any two of its key features. 5

Section B

2 a) Explain the working of asynchronous file reading in Node.js using the `fs` module. Write a program that performs the following tasks: 8

i. Open the file `fruit.txt` asynchronously.
ii. Read its contents in chunks of `5` bytes using a `buffer` and recursively concatenate them to form the complete string.
iii. Display the total contents of the file on the console after all chunks have been read.

b) Explain how `Node.js` handles incoming requests, including the roles of the `event loop`, `event queue`, and `thread pool`, with the help of a diagram. 7

3 a) Explain how a `Node.js` server handles dynamic `GET` requests and generates a response. Write a `Node.js` program that: 8

i. Creates a server listening on port `8080`.
ii. Sends a dynamic `HTML` response containing an array of messages `['Hello World', 'From a basic Node.js server', 'Take Luck']`.
iii. Use `res.write()` to stream the `HTML` and `res.end()` to finalize the response.

b) Describe the role of the Response object in `Express`, including setting `headers, status codes`, and `sending responses`. Implement an Express program to demonstrate this. 7

4 a) Explain how cookies are managed in Express using `cookieParser`, and write a program to set a user cookie with a 1-hour lifespan. 7

b) Explain how to perform `CRUD (Create, Read, Update, Delete)` operations in `MongoDB` using `Node.js`. Write code examples to demonstrate the following tasks: 8

i. Inserting a new document into a collection.

ii. Retrieving documents from the collection.

iii. Updating an existing document.

iv. Deleting a document from the collection.

5 a) Explain the distinctions between `Git, GitHub`, and `GitLab`, outlining the specific roles of each platform. Provide examples demonstrating their application in a standard software development workflow. 8

b) Explain the architecture of `Jenkins`. Describe the roles of the Master node, Agent nodes, and how jobs are executed in this setup. 7

6 a) Compare `Virtual Machines (VMs)` and `Docker containers`. Highlight at least three key differences and their impact on resource utilization. 8

b) Define the `Project Object Model (POM)` in `Maven`. Explain its significance and the key elements it contains. 7

7 a) Discuss the properties of the cookie option in `cookieSession`, including `maxAge, path, httpOnly`, and `signed`. How do these properties affect session management? 7

b) Explain the purpose and usage of `res.json()` and `res.jsonp()` in Express, highlighting their advantages over sending `HTML`. Write `Express` routes to demonstrate how both methods send data to the client. 8